

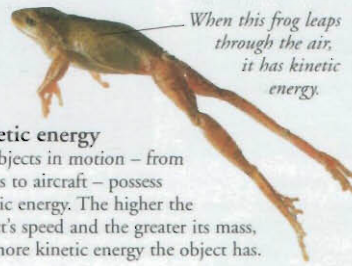
ENERGY



WE RELY ON THE ENERGY stored in food to keep us alive and on the energy locked within fuels to drive our machines and industries. Energy is the ability to make things happen, whether it is moving something, heating it up, or changing it in some way. Energy exists in many different forms, including electricity, sound, heat, and light.

Types of energy

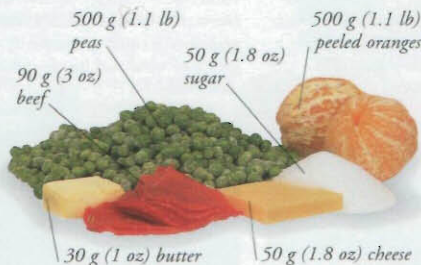
All energy is either kinetic or potential. Kinetic energy is the energy of moving objects, while potential energy is energy that is stored, ready for use. Energy is measured in units called joules (J).



When this frog leaps through the air, it has kinetic energy.

Kinetic energy

All objects in motion – from atoms to aircraft – possess kinetic energy. The higher the object's speed and the greater its mass, the more kinetic energy the object has.

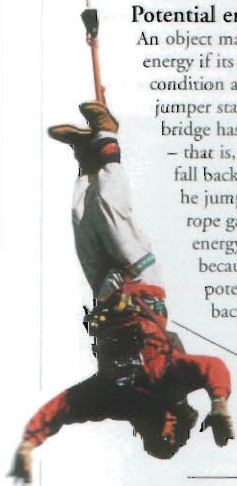


Chemical energy

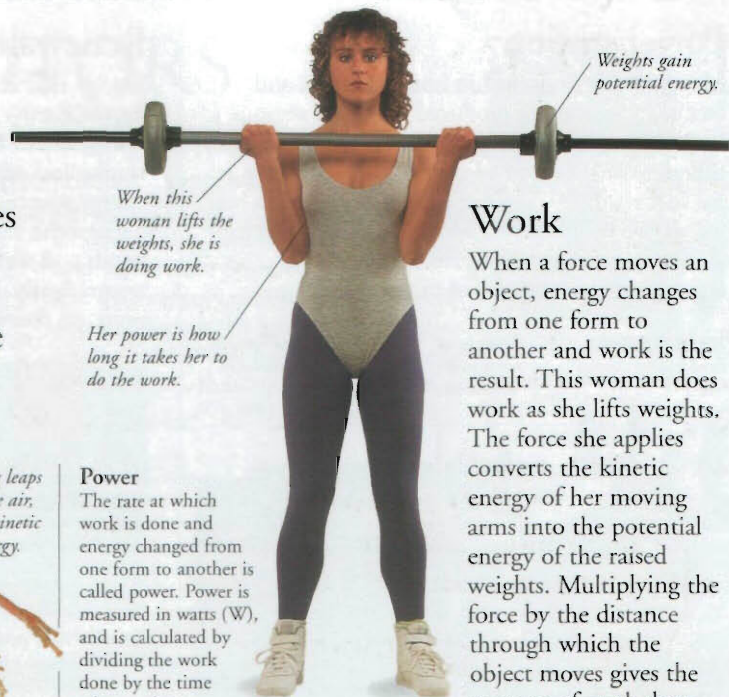
Foods and fuels contain energy stored within chemical compounds. This is a type of potential energy called chemical energy. Some foods store more energy than others. All the foods above contain the same amount of energy, but you would have to eat 500 g (1.1 lb) of peas to get as much energy as you would from just 30 g (1 oz) of butter.

Potential energy

An object may gain potential energy if its position or condition alters. A bungee jumper standing on top of a bridge has potential energy – that is, the potential to fall back to Earth. When he jumps, his bungee rope gains potential energy as it stretches, because it has the potential to pull him back up again.



The jumper's potential energy changes to kinetic energy as he falls.



When this woman lifts the weights, she is doing work.

Her power is how long it takes her to do the work.

Weights gain potential energy.

Work

When a force moves an object, energy changes from one form to another and work is the result. This woman does work as she lifts weights. The force she applies converts the kinetic energy of her moving arms into the potential energy of the raised weights. Multiplying the force by the distance through which the object moves gives the amount of work done.

Power

The rate at which work is done and energy changed from one form to another is called power. Power is measured in watts (W), and is calculated by dividing the work done by the time taken to do it.



Electrical power

Every electrical appliance is given a power rating. If a fan has a power rating of 100 W, it shows that the fan converts 100 J of electrical energy into kinetic energy each second. Similarly, a 1,000 W iron changes electricity into heat at the rate of 1,000 J per second.

Both bulbs give out the same light.



Fluorescent bulb uses less electricity.

Efficiency

Out of every 100 J of electrical energy used by a 60 W incandescent bulb, only 10 J are changed into light; the rest are lost as heat. The bulb has an efficiency of 10%. A 15 W fluorescent bulb is 40% efficient. It gives the same light using a quarter of the electricity.

Energy transfer

The Law of Conservation of Energy says that energy is always conserved – that is, it can be neither created nor destroyed. This law means that when objects gain or lose energy, the energy simply transfers from place to place, or changes into a different form.



1 Tremendous temperatures at the Sun's surface cause it to give out light and other forms of energy, some of which reach the Earth.

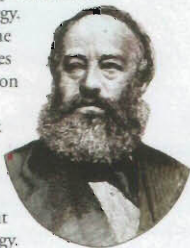
2 When sunlight falls on plants, some of the light energy transfers to the plants by a process called photosynthesis. It is stored as chemical energy.

3 Eating plant-based food, such as bread, enables you to break down the food. This releases the chemical energy and transfers it to your body.

4 Riding a bicycle changes the chemical energy into kinetic energy. If you brake, friction changes this energy into heat as you slow down.

James Joule

The unit of energy, the joule, is named after the English physicist James Joule (1818–89), who helped to develop the Law of Conservation of Energy. Joule noticed that if he rotated a set of paddles in water, the water soon became warm. He realized that the work of turning the paddles changed their kinetic energy into heat, proving that heat is a form of energy.



Timeline

1829 French physicist Gustave Coriolis introduces the term "kinetic energy".

1843 James Joule's experiments show how heat, work, and power are related.

1847 Joule and German physicists Hermann von Helmholtz and Julius Meyer independently state the Law of Conservation of Energy.

1853 Scottish scientist William Rankine devises the concept of "potential energy".



1881 The world's first electricity-generating power station opens in Surrey, UK.

1884 Irish engineer Charles Parsons invents the steam turbine.

Parsons' turbine

1905 German physicist Albert Einstein suggests that matter is a form of energy, and vice versa.

1980s Declining fossil fuel reserves and pollution bring calls for machines and industries to be more energy efficient.